PARDEEP KUMAR

Scientific Software Developer

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SUMMARY

Scientific software engineer with 8+ years of experience in C++, Python, C#, and Julia with cross-industry experience in energy, semiconductors, finance, and automotive software. Experienced in developing reliable, efficient, and maintainable software for high-performance, embedded, and engineering applications. Specialized in high-performance scientific software, numerical methods, and constraint-based algorithm development. Experienced in translating complex mathematical and physical models into robust and efficient code.

EXPERIENCE

Aug 2022 – Present

PhD (Scientific Computing)

Centrum Wiskunde & Informatica, Amsterdam

- Developed simulation tools for multiphase, multicomponent fluid flow in pipelines (Julia, Python) .
- Developed constrained-optimization based thermodynamic phase-equilibrium models and validation workflows for physics-based simulations; 30x faster compared to traditional approach.
- Developed a neural-network-based thermodynamics module to accelerate phase-equilibrium predictions and reduce computational cost in large-scale simulations in Julia and Python.
- Developed an electromagnetics solver for 3D unstructured mesh using Finite volume time domain method (C++, Julia).
- Translated a Python-based solver for Einstein's equations of general relativity into Julia, and extended the model with numerical methods commonly used in computational fluid dynamics.
- Published work in peer-reviewed journals.

Feb 2022 – Aug 2022

Software Engineer

ASM International, Almere

- Worked on the development of a Java-based tool simulator for semiconductor film deposition tools (hardware): implemented feature enhancements to improve functionality and usability for internal engineering workflows (Java).
- Refactored legacy Java code to improve modularity and maintainability, while fixing bugs and supporting incremental feature development.
- Received hands-on training on multiple modules of film deposition equipment, developing a practical understanding of semiconductor tool architecture, hardware modules and operational workflows.

Jun 2019 – Dec 2021

Software Engineer

Shell, Amsterdam

- Migrated solver workflows from Python to C++/CUDA, improving performance through GPU parallelization and reduced the runtime from 30 minutes to 15 seconds.
- Designed and developed a framework for efficient retrieval and management of data from server using a concurrent algorithm for multiple instances of a tool which reads and writes data back and forth between real-time field databases (PI from OSI) and process and pipeline simulation software (UniSim Design and COMPAS). The performance of the application was greatly enhanced (15+ times faster).
- Built high-performance concurrent frameworks for server data access, improving throughput for multiple clients using C#, Managed C++, and C++.
- Migrated APIs with a Managed C++ interoperability layer between C# and C++.
- Developed a modular data visualization library for the Shell in-house dynamic multi-phase flow simulator COMPAS.

Feb 2019 – Apr 2019

Software Engineer

Aakraya Research, Bangalore

- Developed offline-regression models generator from historical market data for High Frequency Trading (Python).
- Developed application to generate profit and loss metrics for analyzing the performance of the models after live trading (Python).
- Improved the orderbook for efficient order matching (C++).

May 2018 – Nov 2018

Software Engineer II

KLA-Tencor, Chennai, India

- Built server framework infrastructure for cross-application integration using C# and F#.
- Designed a silica wafer thickness simulation engine using C#, F#, and MATLAB.
- Developed libraries for common functionalities for server apps (C++, C#).
- Implemented a fault-tolerance system in Erlang with seamless failover and recovery.
- Refactored legacy code into a more modular architecture, fixed bugs, implemented feature enhancements, and developed unit tests.

Mar 2016 – Apr 2018

Software Engineer

Altair, Bangalore, India

- Developed shared-memory infrastructure for high-volume data transfer across processes (C++).
- Developed various Geometry and Mesh manipulation tools for Hypermesh (meshing tool).
- Migrated bindings from SWIG to Boost.Python to improve C++/Python interoperability.
- Refactored the code to make the application (Automate: Simulink type application) Model View Controller compliant (C++) and features enhancement.

- Designed unit testing frameworks in Python for C++ entities and extended COM-based APIs.

Jul 2014 – Mar 2016

Research & Development Engineer

Fluidyn, Bangalore, India

- Developed a Finite Volume Time Domain electromagnetic solver with MPI parallelization using C++.
- Implemented various numerical schemes for spatial 2nd and 3rd order spatial accuracy.
- Developed library to compute near to far field transformation to predict Radar Cross Section of stealth aircraft (C++, MPI).
- Enhanced the fluid solver (Navier-Stokes) to compute far-field noise by performing Retarded time integration on a near field surface data (C++).

TECHNICAL SKILLS

Programming
C++, Python, C#, Julia, F#, MATLAB, Java

Simulation & modeling
Numerical methods, optimization, linear algebra, CFD, computational electromagnetics, FEM, Finite differences

High performance computing
Concurrent programming, GPU acceleration, MPI parallelization, shared memory

AI & data pipelines
Neural-network thermodynamics, training & validation automation, data workflow design

Tools & practices
Git, Perforce, test-driven development, unit testing, SDLC

Software architecture
Object-oriented programming, functional programming, component object model

EDUCATION

PhD (2022 - 2026)

Mechanical Engineering, TU Delft, The Netherlands

MSc (2012 - 2014)

Aerospace Engineering, IIT Bombay, India

LANGUAGES

Proficient
English and Hindi

Beginner (A2)
Dutch